

EXPERIMENT 19: Prolog Program for Student, Teacher and Subject

Aim

To write and implement a **Prolog program to represent the relationship between students, teachers, and subjects** using facts and rules.

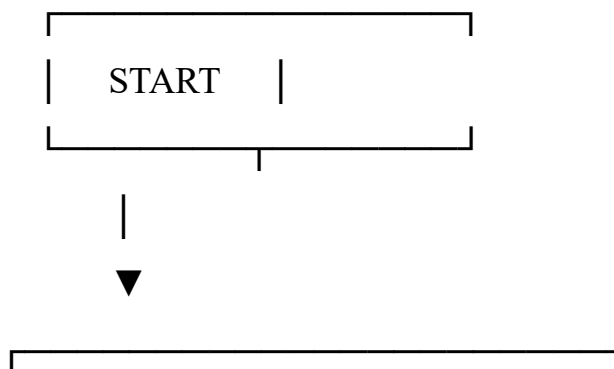
Objective

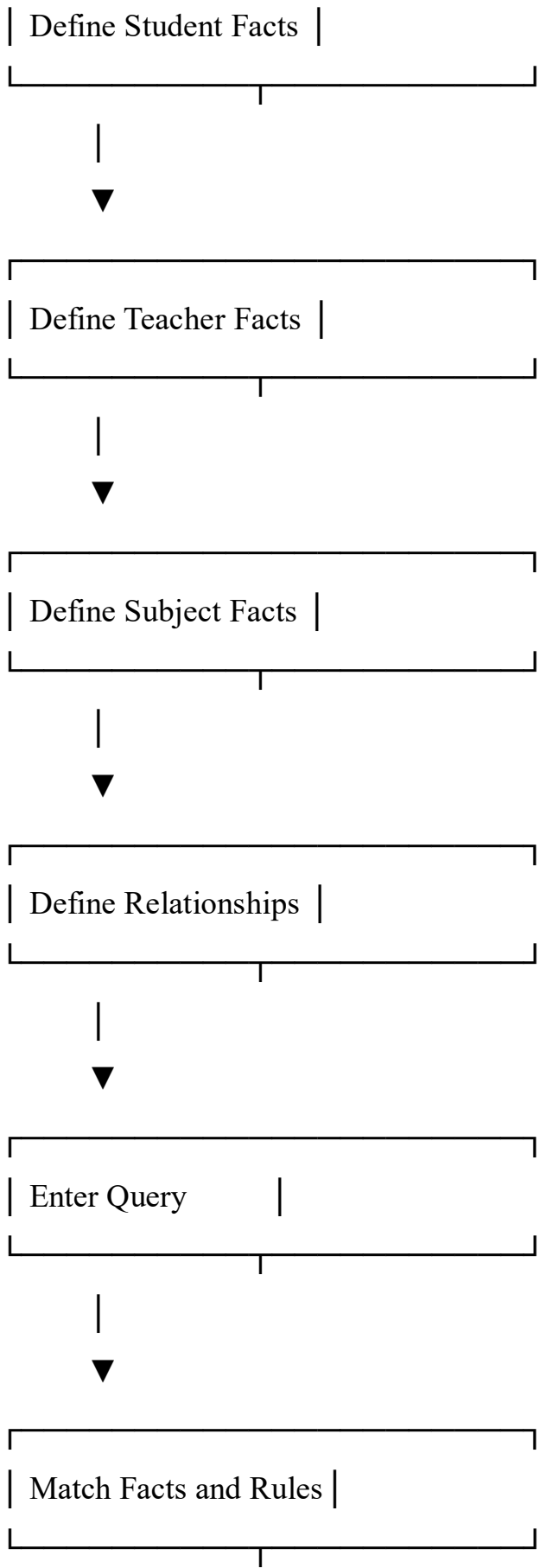
- To understand facts and rules in Prolog.
- To represent student, teacher, and subject information.
- To find which teacher teaches a particular subject.
- To find which students study a particular subject.

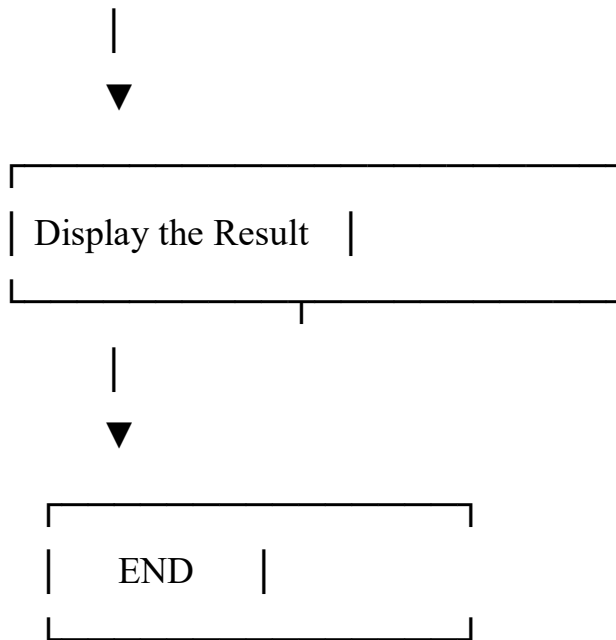
Algorithm

5. Start the program.
6. Define facts for students.
7. Define facts for teachers.
8. Define facts for subjects.
9. Define relationships between teachers and subjects.
10. Define relationships between students and subjects.
11. Create rules to find the required relationships.
12. Execute queries to retrieve information.
13. Display the result.
14. Stop the program.

Flowchart







Code

% Prolog Program for Student, Teacher and Subject

% Student facts

student(rahul).

student(priya).

student(amit).

student(sneha).

% Teacher facts

teacher(rajesh).

teacher(suresh).

teacher(anita).

% Subject facts

subject(math).

subject(science).

```
subject(computer).
```

```
% Teacher teaches Subject
```

```
teaches(rajesh, math).
```

```
teaches(suresh, science).
```

```
teaches(anita, computer).
```

```
% Student studies Subject
```

```
studies(rahul, math).
```

```
studies(rahul, computer).
```

```
studies(priya, science).
```

```
studies(priya, math).
```

```
studies(amit, computer).
```

```
studies(sneha, science).
```

```
% Rule to find teacher of a student's subject
```

```
student_teacher(Student, Teacher) :-
```

```
    studies(Student, Subject),
```

```
    teaches(Teacher, Subject).
```

Queries

1. Find the subjects studied by Rahul

```
?- studies(rahul, Subject).
```

Output

```
Subject = math ;
```

```
Subject = computer.
```

2. Find the teacher who teaches Math

```
?- teaches(Teacher, math).
```

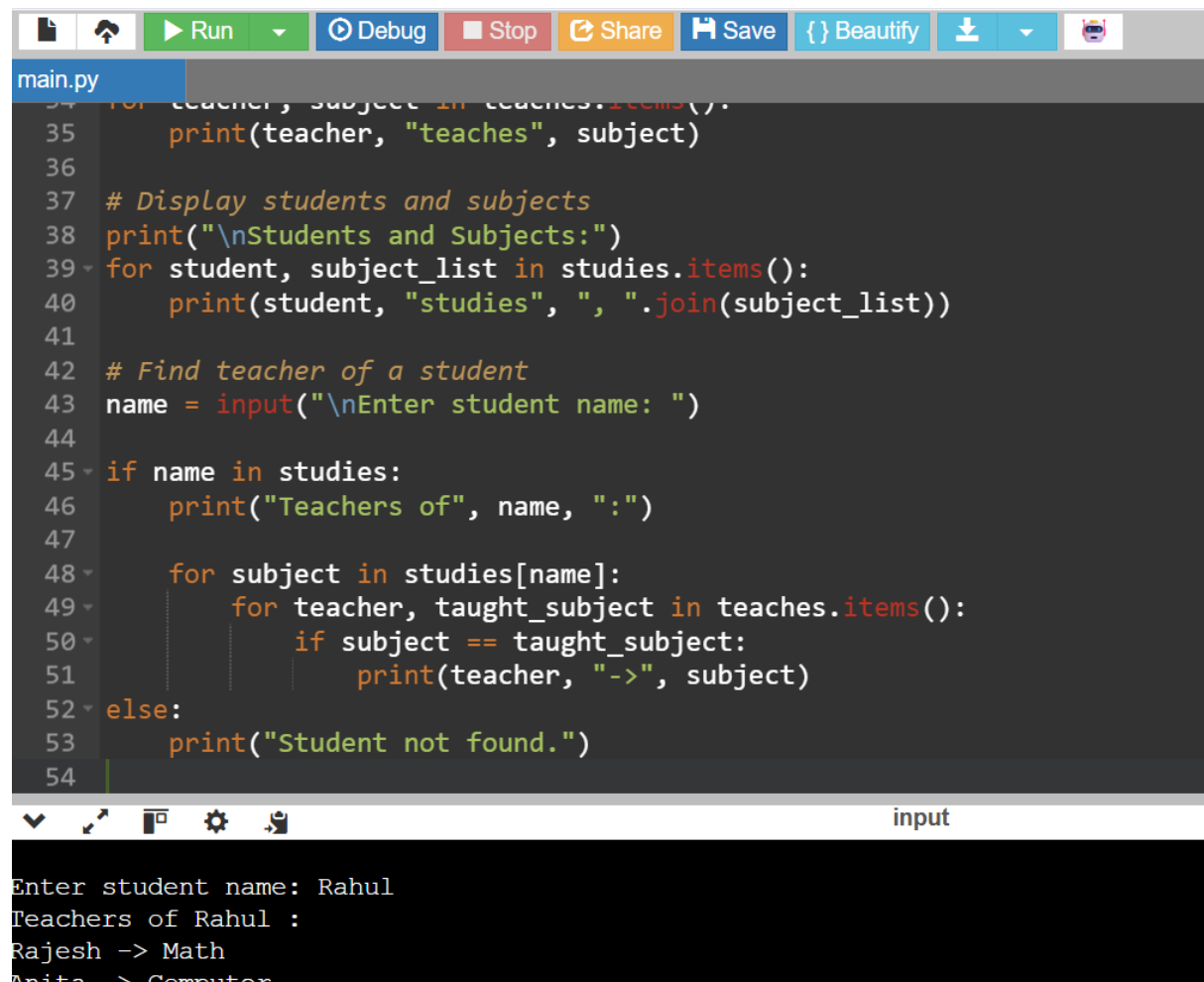
Output

Teacher = rajesh.

3. Find the teacher of Rahul

?- student_teacher(rahul, Teacher).

Output



The screenshot shows a Python IDE with a toolbar at the top containing buttons for Run, Debug, Stop, Share, Save, Beautify, and a download icon. The main editor window displays a Python script named 'main.py'. The script defines two dictionaries: 'teaches' mapping teachers to subjects, and 'studies' mapping students to lists of subjects. It includes functions to print these relationships and a function to find the teacher of a given student. The output window at the bottom shows the program's execution: it prompts for a student name, receives 'Rahul', and prints the teachers 'Rajesh' (for Math) and 'Anita' (for Computer).

```
main.py
34 for teacher, subject in teaches.items():
35     print(teacher, "teaches", subject)
36
37 # Display students and subjects
38 print("\nStudents and Subjects:")
39 for student, subject_list in studies.items():
40     print(student, "studies", ", ".join(subject_list))
41
42 # Find teacher of a student
43 name = input("\nEnter student name: ")
44
45 if name in studies:
46     print("Teachers of", name, ":")
47
48     for subject in studies[name]:
49         for teacher, taught_subject in teaches.items():
50             if subject == taught_subject:
51                 print(teacher, "->", subject)
52 else:
53     print("Student not found.")
54
```

input

Enter student name: Rahul
Teachers of Rahul :
Rajesh -> Math
Anita -> Computer

Teacher = rajesh ;

Teacher = anita.

Result

The Prolog program was successfully implemented to represent and retrieve relationships between **students, teachers, and subjects**.

Conclusion

The program demonstrates the use of **facts, rules, variables, and queries in Prolog**. It can determine which subjects a student studies and which teacher teaches those subjects.